

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
#df = pd.read_csv("Titanic-Dataset.csv")
#print(df.describe())
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
data = pd.read_csv("Titanic-Dataset.csv")
```

```
data.head()
```

	PassengerId	Survived	Pclass \
0	1	0	3
1	2	1	1
2	3	1	3
3	4	1	1
4	5	0	3

	Name	Sex	Age
SibSp \			
0	Braund, Mr. Owen Harris	male	22.0
1			
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1			
2	Heikkinen, Miss. Laina	female	26.0
0			
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
1			
4	Allen, Mr. William Henry	male	35.0

0

	Parch		Ticket	Fare	Cabin	Embarked
0	0		A/5 21171	7.2500	NaN	S
1	0		PC 17599	71.2833	C85	C
2	0	STON/O2.	3101282	7.9250	NaN	S
3	0		113803	53.1000	C123	S
4	0		373450	8.0500	NaN	S

data.tail()

	PassengerId	Survived	Pclass	
Name \				
886	887	0	2	Montvila, Rev. Juozas
887	888	1	1	Graham, Miss. Margaret Edith
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"
889	890	1	1	Behr, Mr. Karl Howell
890	891	0	3	Dooley, Mr. Patrick

	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	male	27.0	0	0	211536	13.00	NaN	S
887	female	19.0	0	0	112053	30.00	B42	S
888	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	male	26.0	0	0	111369	30.00	C148	C
890	male	32.0	0	0	370376	7.75	NaN	Q

data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass          891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp           891 non-null   int64
7   Parch           891 non-null   int64
8   Ticket          891 non-null   object
9   Fare           891 non-null   float64
10  Cabin           204 non-null   object
11  Embarked        889 non-null   object
```

dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

data.dtypes

PassengerId int64
Survived int64
Pclass int64
Name object
Sex object
Age float64
SibSp int64
Parch int64
Ticket object
Fare float64
Cabin object
Embarked object
dtype: object

data.nunique

<bound method DataFrame.nunique of				PassengerId	Survived	Pclass
\						
0	1	0	3			
1	2	1	1			
2	3	1	3			
3	4	1	1			
4	5	0	3			
..	...	...	...			
886	887	0	2			
887	888	1	1			
888	889	0	3			
889	890	1	1			
890	891	0	3			

		Name	Sex	Age
SibSp	\			
0		Braund, Mr. Owen Harris	male	22.0
1				
1		Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1				
2		Heikkinen, Miss. Laina	female	26.0
0				
3		Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
1				
4		Allen, Mr. William Henry	male	35.0
0				
..		...	...	...
...				
886		Montvila, Rev. Juozas	male	27.0

```

0
887          Graham, Miss. Margaret Edith  female  19.0
0
888          Johnston, Miss. Catherine Helen "Carrie"  female  NaN
1
889          Behr, Mr. Karl Howell  male  26.0
0
890          Dooley, Mr. Patrick  male  32.0
0

```

```

      Parch      Ticket      Fare Cabin Embarked
0         0      A/5 21171   7.2500   NaN      S
1         0      PC 17599  71.2833   C85      C
2         0  STON/O2. 3101282   7.9250   NaN      S
3         0      113803  53.1000  C123      S
4         0      373450   8.0500   NaN      S
..      ...      ...      ...      ...      ...
886        0      211536  13.0000   NaN      S
887        0      112053  30.0000   B42      S
888        2      W./C. 6607  23.4500   NaN      S
889        0      111369  30.0000  C148      C
890        0      370376   7.7500   NaN      Q

```

```
[891 rows x 12 columns]>
```

```
data.isnull
```

```

<bound method DataFrame.isnull of      PassengerId  Survived
Pclass \
0          1          0          3
1          2          1          1
2          3          1          3
3          4          1          1
4          5          0          3
..      ...      ...      ...
886      887          0          2
887      888          1          1
888      889          0          3
889      890          1          1
890      891          0          3

```

```

      Name      Sex  Age
SibSp \
0      Braund, Mr. Owen Harris  male  22.0
1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0
1
2      Heikkinen, Miss. Laina  female  26.0
0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0

```

```

1
4 Allen, Mr. William Henry male 35.0
0
.. ... ...
...
886 Montvila, Rev. Juozas male 27.0
0
887 Graham, Miss. Margaret Edith female 19.0
0
888 Johnston, Miss. Catherine Helen "Carrie" female NaN
1
889 Behr, Mr. Karl Howell male 26.0
0
890 Dooley, Mr. Patrick male 32.0
0

```

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S
..	...	...	...	...	...
886	0	211536	13.0000	NaN	S
887	0	112053	30.0000	B42	S
888	2	W./C. 6607	23.4500	NaN	S
889	0	111369	30.0000	C148	C
890	0	370376	7.7500	NaN	Q

```
[891 rows x 12 columns]>
```

```
data.isnull().sum()
```

```

PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64

```

```
(data.isnull().sum()/(len(data)))*100
```

```
PassengerId    0.000000
Survived       0.000000
Pclass         0.000000
Name           0.000000
Sex            0.000000
Age           19.865320
SibSp          0.000000
Parch          0.000000
Ticket         0.000000
Fare           0.000000
Cabin          77.104377
Embarked       0.224467
dtype: float64
```

```
# Remove S.No. column from data
```

```
print(data.columns)
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age',
      'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

```
data = data.drop(columns=['PassengerId'])
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 11 columns):
```

#	Column	Non-Null Count	Dtype
0	Survived	891 non-null	int64
1	Pclass	891 non-null	int64
2	Name	891 non-null	object
3	Sex	891 non-null	object
4	Age	714 non-null	float64
5	SibSp	891 non-null	int64
6	Parch	891 non-null	int64
7	Ticket	891 non-null	object
8	Fare	891 non-null	float64
9	Cabin	204 non-null	object
10	Embarked	889 non-null	object

```
dtypes: float64(2), int64(4), object(5)
```

```
memory usage: 76.7+ KB
```

```
# Recreate PassengerId (1 to 891)
```

```
data['PassengerId'] = range(1, len(data) + 1)
```

```
# Move PassengerId to the first column
```

```
cols = data.columns.tolist()
```

```
cols = ['PassengerId'] + [col for col in cols if col != 'PassengerId']
data = data[cols]
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

```
dtypes: float64(2), int64(5), object(5)
```

```
memory usage: 83.7+ KB
```

```
%pip install seaborn
```

```
Collecting seaborn
```

```
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
```

```
Requirement already satisfied: numpy!=1.24.0,>=1.20 in d:\college\machine learning (ml)\.venv\lib\site-packages (from seaborn) (2.3.1)
```

```
Requirement already satisfied: pandas>=1.2 in d:\college\machine learning (ml)\.venv\lib\site-packages (from seaborn) (2.3.1)
```

```
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in d:\college\machine learning (ml)\.venv\lib\site-packages (from seaborn) (3.10.3)
```

```
Requirement already satisfied: contourpy>=1.0.1 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.2)
```

```
Requirement already satisfied: cycler>=0.10 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)
```

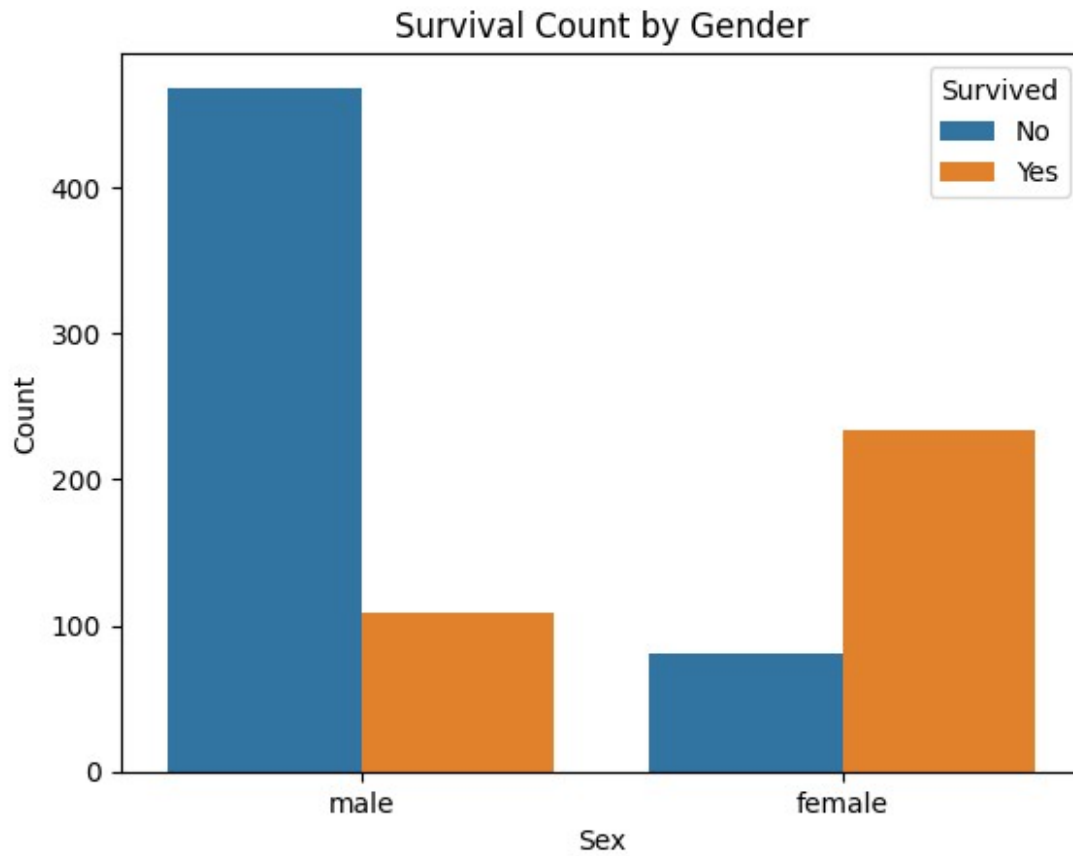
```
Requirement already satisfied: fonttools>=4.22.0 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.59.0)
```

```
Requirement already satisfied: kiwisolver>=1.3.1 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.8)
```

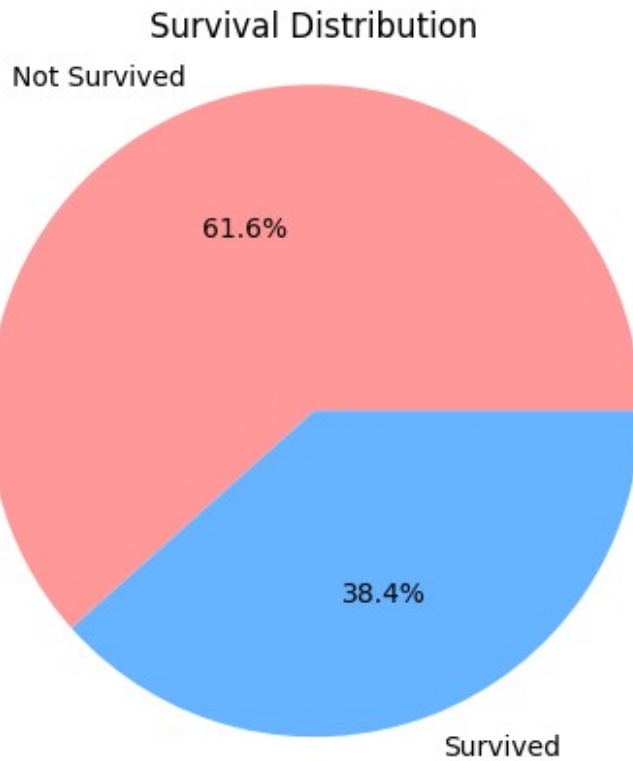
```
Requirement already satisfied: packaging>=20.0 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (25.0)
```

Requirement already satisfied: pillow>=8 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in d:\college\machine learning (ml)\.venv\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in d:\college\machine learning (ml)\.venv\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in d:\college\machine learning (ml)\.venv\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Installing collected packages: seaborn
Successfully installed seaborn-0.13.2
Note: you may need to restart the kernel to use updated packages.

```
import matplotlib.pyplot as plt
import seaborn as sns
# Bar chart: Survived vs Sex
sns.countplot(x='Sex', hue='Survived', data=data)
plt.title('Survival Count by Gender')
plt.xlabel('Sex')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```

```
# Pie chart: Overall survival
survived_counts = data['Survived'].value_counts()
labels = ['Not Survived', 'Survived']
plt.pie(survived_counts, labels=labels, autopct='%1.1f%%',
        colors=['#ff9999', '#66b3ff'])
plt.title('Survival Distribution')
plt.axis('equal')
plt.show()
```



```
# Histogram: Age distribution  
plt.hist(data['Age'].dropna(), bins=20, color='skyblue',  
edgecolor='black')  
plt.title('Age Distribution of Passengers')  
plt.xlabel('Age')  
plt.ylabel('Frequency')  
plt.show()
```

